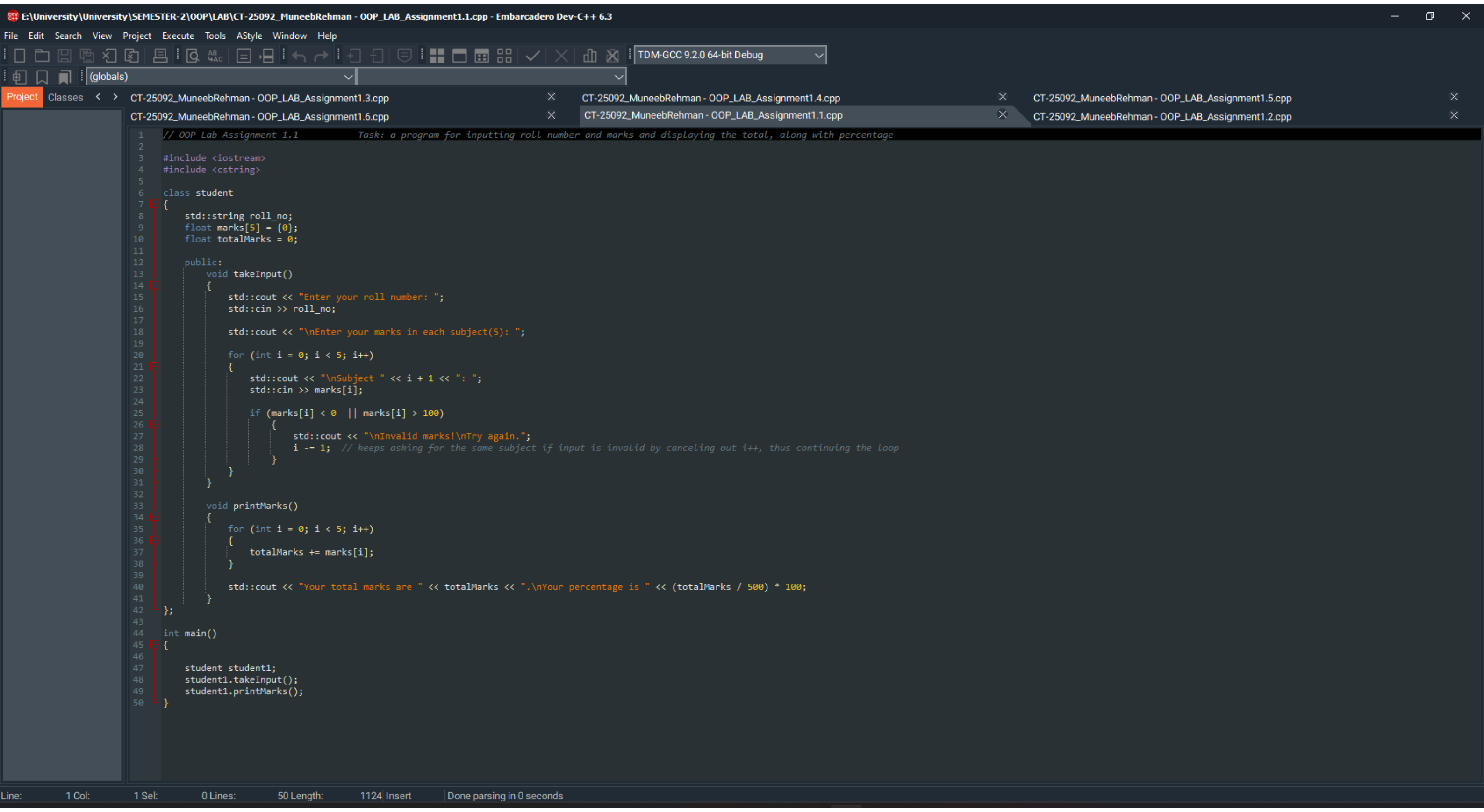
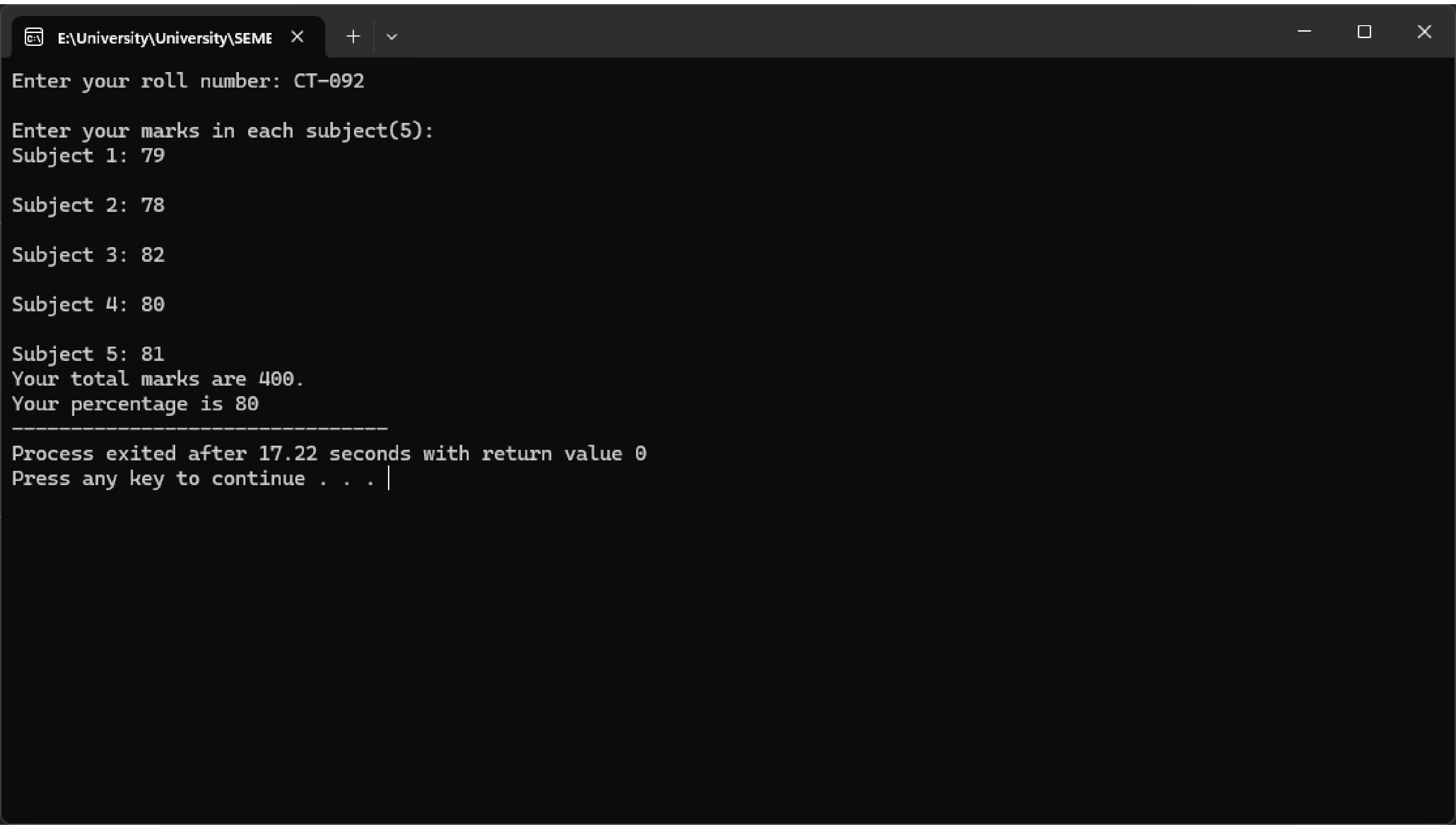
Muneeb Rehman, CT-092 Assignment #1

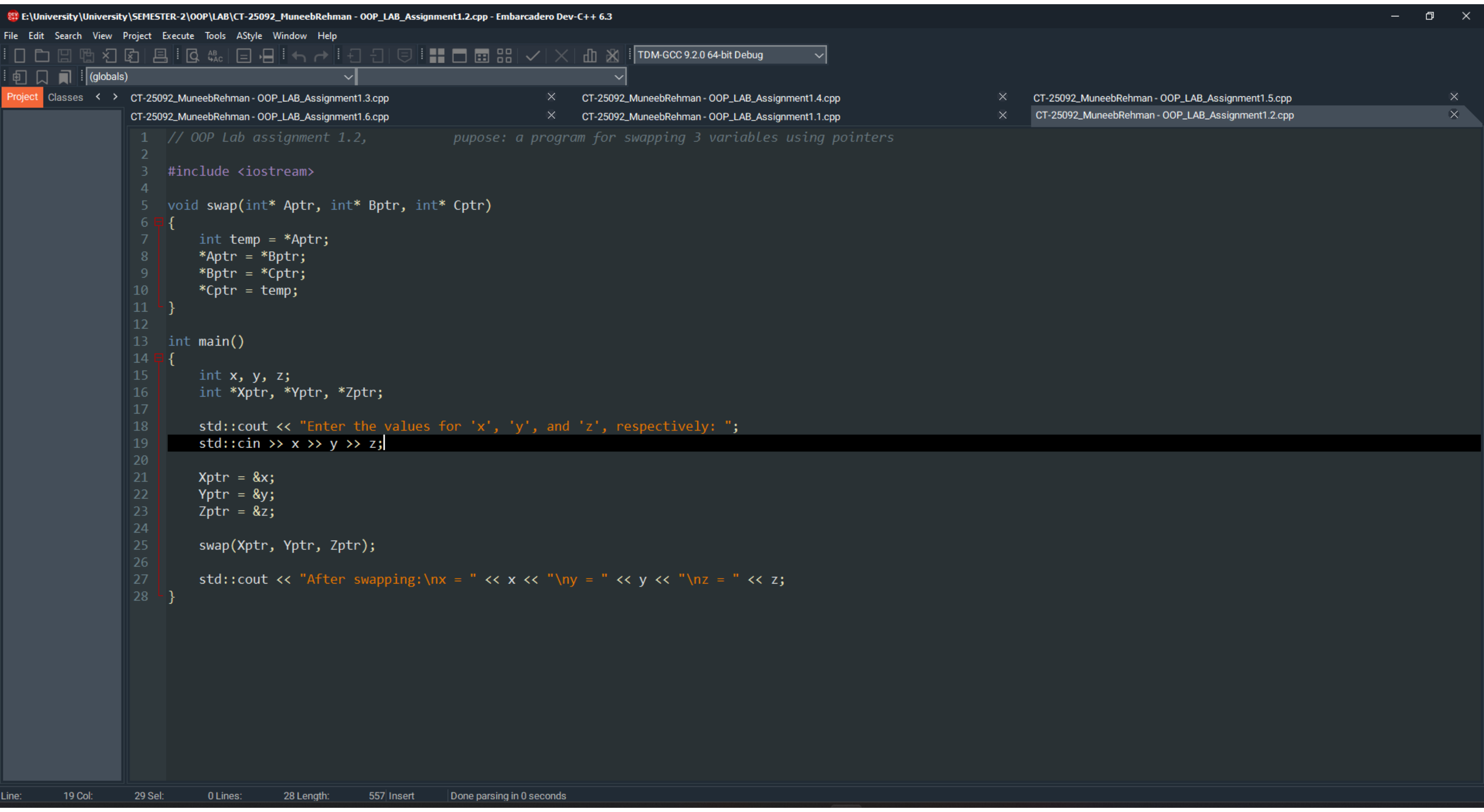
Question 1.






```
Enter your roll number: CT-092

Enter your marks in each subject(5):
Subject 1: 79

Subject 2: 78

Subject 3: 82

Subject 4: 80

Subject 5: 81
Your total marks are 400.
Your percentage is 80
--------------------------------
Process exited after 17.22 seconds with return value 0
Press any key to continue . . .
```

Question 2.


E:\University\University\SEMESTER-2\OOP\LAB\CT-25092_MuneebRehman - OOP_LAB_Assignment1.2.cpp - Embarcadero Dev-C++ 6.3
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 9.2.0 64-bit Debug
(globals)
Project Classes
CT-25092_MuneebRehman - OOP_LAB_Assignment1.3.cpp
CT-25092_MuneebRehman - OOP_LAB_Assignment1.4.cpp
CT-25092_MuneebRehman - OOP_LAB_Assignment1.5.cpp
CT-25092_MuneebRehman - OOP_LAB_Assignment1.6.cpp
CT-25092_MuneebRehman - OOP_LAB_Assignment1.1.cpp
CT-25092_MuneebRehman - OOP_LAB_Assignment1.2.cpp
// OOP Lab assignment 1.2,   pupose: a program for swapping 3 variables using pointers

#include <iostream>

void swap(int* Aptr, int* Bptr, int* Cptr)
{
    int temp = *Aptr;
    *Aptr = *Bptr;
    *Bptr = *Cptr;
    *Cptr = temp;
}

int main()
{
    int x, y, z;
    int *Xptr, *Yptr, *Zptr;

    std::cout << "Enter the values for 'x', 'y', and 'z', respectively: ";
    std::cin >> x >> y >> z;

    Xptr = &x;
    Yptr = &y;
    Zptr = &z;

    swap(Xptr, Yptr, Zptr);

    std::cout << "After swapping:\nx = " << x << "\ny = " << y << "\nz = " << z;
}
Line: 19 Col: 29 Sel: 0 Lines: 28 Length: 557 Insert Done parsing in 0 seconds

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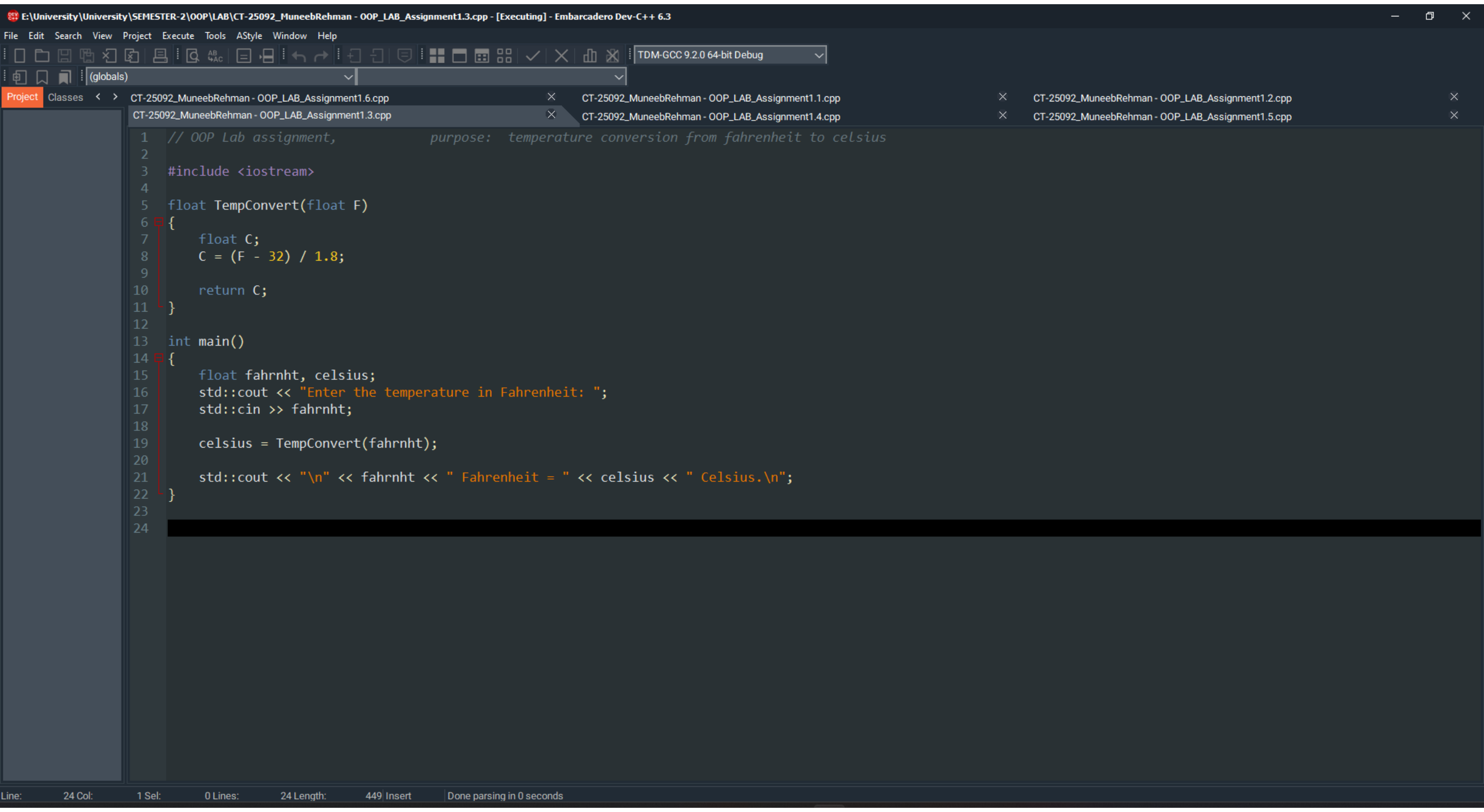
```
Enter the values for 'x', 'y', and 'z', respectively: 12
15
19
After swapping:
x = 15
y = 19
z = 12
--------------------------------
Process exited after 12.65 seconds with return value 0
Press any key to continue . . .
```

Question 3


E:\University\University\SEMESTER-2\OOP\LAB\CT-25092_MuneebRehman - OOP_LAB_Assignment1.3.cpp - [Executing] - Embarcadero Dev-C++ 6.3
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 9.2.0 64-bit Debug
(globals)
Project Classes
CT-25092_MuneebRehman - OOP_LAB_Assignment1.6.cpp
CT-25092_MuneebRehman - OOP_LAB_Assignment1.1.cpp
CT-25092_MuneebRehman - OOP_LAB_Assignment1.2.cpp
CT-25092_MuneebRehman - OOP_LAB_Assignment1.3.cpp
CT-25092_MuneebRehman - OOP_LAB_Assignment1.4.cpp
CT-25092_MuneebRehman - OOP_LAB_Assignment1.5.cpp
// OOP Lab assignment,   purpose: temperature conversion from fahrenheit to celsius

#include <iostream>

float TempConvert(float F)
{
    float C;
    C = (F - 32) / 1.8;

    return C;
}

int main()
{
    float fahrnht, celsius;
    std::cout << "Enter the temperature in Fahrenheit: ";
    std::cin >> fahrnht;

    celsius = TempConvert(fahrnht);

    std::cout << "\n" << fahrnht << " Fahrenheit = " << celsius << " Celsius.\n";
}
Line: 24 Col: 1 Sel: 0 Lines: 24 Length: 449 Insert Done parsing in 0 seconds

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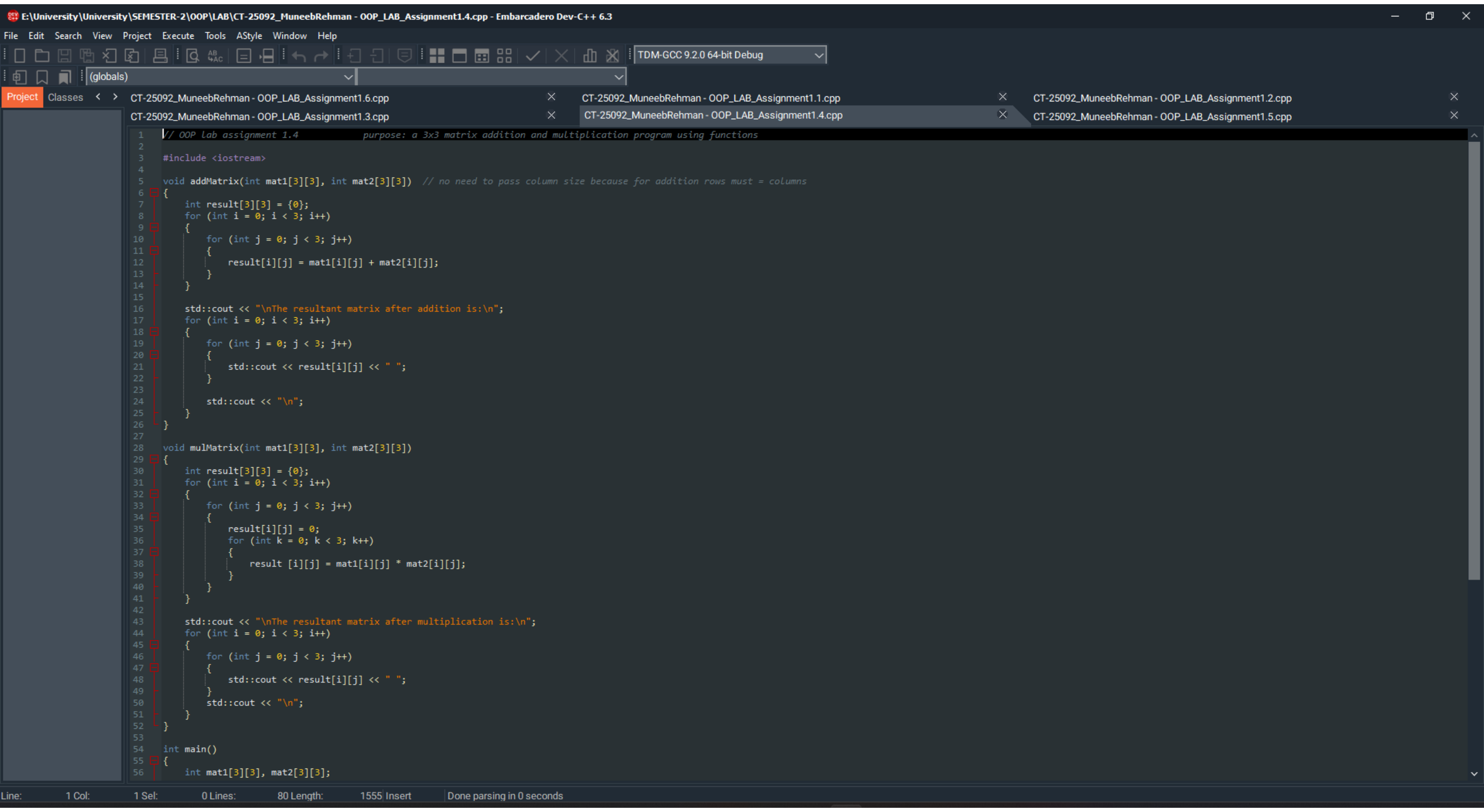
```
Enter the temperature in Fahrenheit: 101.76

101.76 Fahrenheit = 38.7556 Celsius.

--------------------------------
Process exited after 7.204 seconds with return value 0
Press any key to continue . . .
```

Question 4




```cpp
// OOP lab assignment 1.4          purpose: a 3x3 matrix addition and multiplication program using functions

#include <iostream>

void addMatrix(int mat1[3][3], int mat2[3][3])  // no need to pass column size because for addition rows must = columns
{
    int result[3][3] = {0};
    for (int i = 0; i < 3; i++)
    {
        for (int j = 0; j < 3; j++)
        {
            result[i][j] = mat1[i][j] + mat2[i][j];
        }
    }

    std::cout << "\nThe resultant matrix after addition is:\n";
    for (int i = 0; i < 3; i++)
    {
        for (int j = 0; j < 3; j++)
        {
            std::cout << result[i][j] << " ";
        }

        std::cout << "\n";
    }
}

void mulMatrix(int mat1[3][3], int mat2[3][3])
{
    int result[3][3] = {0};
    for (int i = 0; i < 3; i++)
    {
        for (int j = 0; j < 3; j++)
        {
            result[i][j] = 0;
            for (int k = 0; k < 3; k++)
            {
                result [i][j] = mat1[i][j] * mat2[i][j];
            }
        }
    }

    std::cout << "\nThe resultant matrix after multiplication is:\n";
    for (int i = 0; i < 3; i++)
    {
        for (int j = 0; j < 3; j++)
        {
            std::cout << result[i][j] << " ";
        }
        std::cout << "\n";
    }
}

int main()
{
    int mat1[3][3], mat2[3][3];
```

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CT-25092_MuneebRehman - OOP_LAB_Assignment1.6.cpp | CT-25092_MuneebRehman - OOP_LAB_Assignment1.1.cpp | CT-25092_MuneebRehman - OOP_LAB_Assignment1.2.cpp | CT-25092_MuneebRehman - OOP_LAB_Assignment1.3.cpp | CT-25092_MuneebRehman - OOP_LAB_Assignment1.4.cpp | CT-25092_MuneebRehman - OOP_LAB_Assignment1.5.cpp

```cpp
    }
}

void mulMatrix(int mat1[3][3], int mat2[3][3])
{
    int result[3][3] = {0};
    for (int i = 0; i < 3; i++)
    {
        for (int j = 0; j < 3; j++)
        {
            result[i][j] = 0;
            for (int k = 0; k < 3; k++)
            {
                result [i][j] = mat1[i][j] * mat2[i][j];
            }
        }
    }

    std::cout << "\nThe resultant matrix after multiplication is:\n";
    for (int i = 0; i < 3; i++)
    {
        for (int j = 0; j < 3; j++)
        {
            std::cout << result[i][j] << " ";
        }
        std::cout << "\n";
    }
}

int main()
{
    int mat1[3][3], mat2[3][3];

    std::cout << "Enter the elements for the first matrix: ";
    for (int i = 0; i < 3; i++)
    {
        for (int j = 0; j < 3; j++)
        {
            std::cin >> mat1[i][j];
        }
    }


    std::cout << "\nEnter the elements for the second matrix: ";
    for (int i = 0; i < 3; i++)
    {
        for (int j = 0; j < 3; j++)
        {
            std::cin >> mat2[i][j];
        }
    }

    addMatrix(mat1, mat2);

    mulMatrix(mat1, mat2);
}
```

Line: 1 Col: 1 Sel: 0 Lines: 80 Length: 1555 Insert Done parsing in 0 seconds

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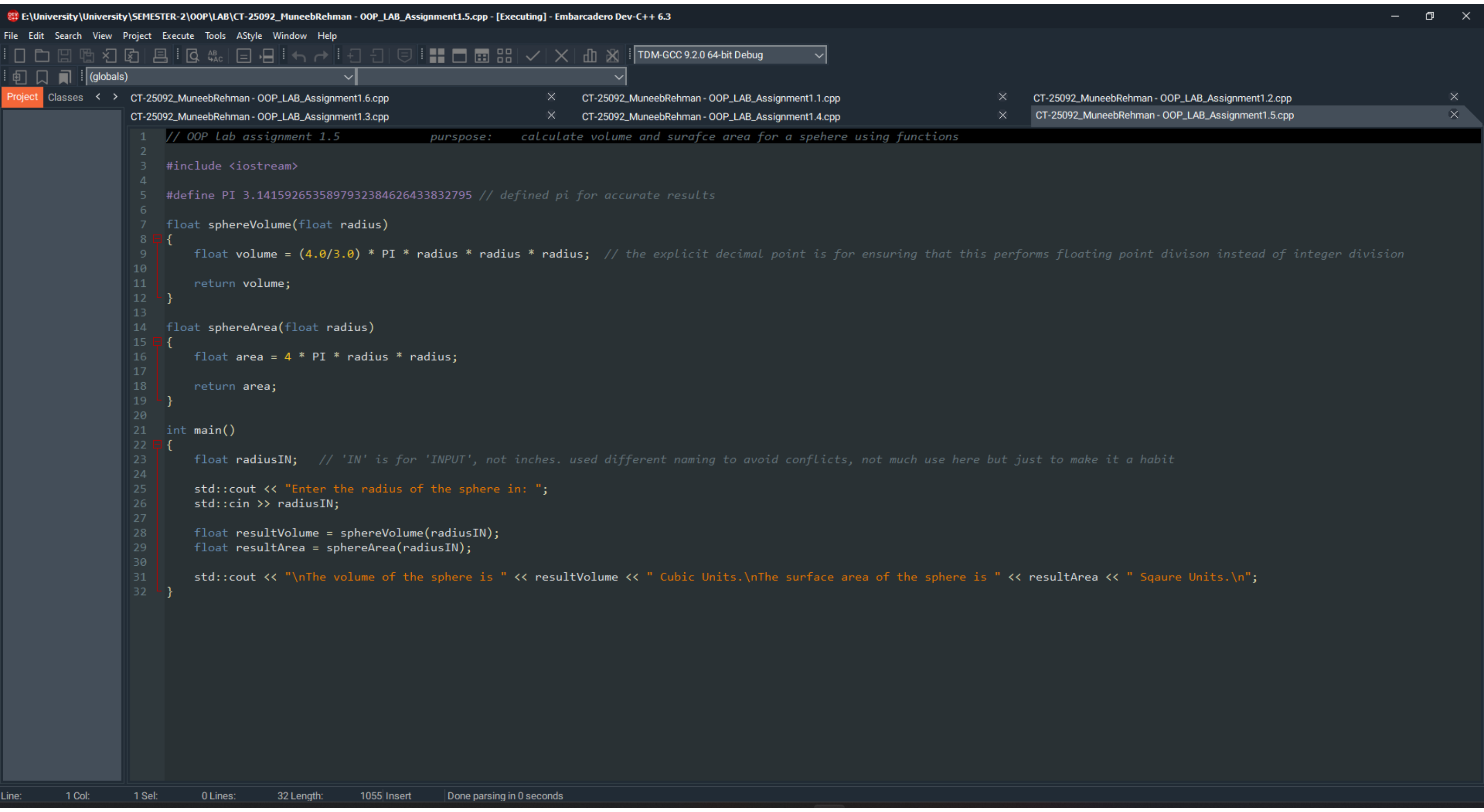
```
Enter the elements for the first matrix: 1 2 3 4 5 6 7 8 9

Enter the elements for the second matrix: 2 4 6 8 10 12 14 16 18

The resultant matrix after addition is:
3 6 9
12 15 18
21 24 27

The resultant matrix after multiplication is:
2 8 18
32 50 72
98 128 162

--------------------------------
Process exited after 29.99 seconds with return value 0
Press any key to continue . . .
```

Question 5.



```cpp
// OOP lab assignment 1.5          pursuose:    calculate volume and surafce area for a sphere using functions

#include <iostream>

#define PI 3.14159265358979323846264338327950 // defined pi for accurate results

float sphereVolume(float radius)
{
    float volume = (4.0/3.0) * PI * radius * radius * radius;  // the explicit decimal point is for ensuring that this performs floating point divison instead of integer division

    return volume;
}

float sphereArea(float radius)
{
    float area = 4 * PI * radius * radius;

    return area;
}

int main()
{
    float radiusIN;   // 'IN' is for 'INPUT', not inches. used different naming to avoid conflicts, not much use here but just to make it a habit

    std::cout << "Enter the radius of the sphere in: ";
    std::cin >> radiusIN;

    float resultVolume = sphereVolume(radiusIN);
    float resultArea = sphereArea(radiusIN);

    std::cout << "\nThe volume of the sphere is " << resultVolume << " Cubic Units.\nThe surface area of the sphere is " << resultArea << " Sqaure Units.\n";
}
```

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```
Enter the radius of the sphere in: 5.789

The volume of the sphere is 812.642 Cubic Units.
The surface area of the sphere is 421.131 Sqaure Units.

--------------------------------
Process exited after 6.377 seconds with return value 0
Press any key to continue . . .
```

Question 6.

```cpp
// OOP assihnment 1.6                    purpose:    Bank cash withdrawal system

#include <iostream>
#include <string>

class bankAccount
{
    private:
        std::string acc_no;   // account no
        int acc_pin;     // account pin
        char acc_type;
        float balance; // account balance


    public:
        bankAccount(std::string accNo, int accPin, char accType)
        {
            acc_no = accNo;
            acc_pin = accPin;
            acc_type = accType;
            balance = 200000;   // default balance of 200k
        }

        void withdraw(float amount)
        {
            if (amount > 100000)
            {
                std::cout << "\nAmount cannot exceed 100,000!";
                return;
            }

            if (acc_type == 'S' || acc_type == 's')
            {
                if (amount <= 50000)
                {
                    balance -= amount - (amount * 0.02);   // 2% transaction fee
                }

                else if (amount > 50000)
                {
                    balance -= amount - (amount * 0.05);    // 5% transaction fee for withdrawal above 50k
                }
            }

            else if (acc_type == 'C' || acc_type == 'c')
            {
                if (amount <= 50000)
                {
                    balance -= amount - 100;   //  100 transaction fee
                }

                else if (amount > 50000)
                {
                    balance -= amount - (amount * 0.05);    // 5% transaction fee for withdrawal above 50k
                }
            }

            std::cout << "\nTransaction successful!\nPlease, take your card and cash.";
            std::cout << "\nYour new balance is " << balance;

        }
```

```cpp
            }

            std::cout << "\nTransaction successful!\nPlease, take your card and cash.";
            std::cout << "\nYour new balance is " << balance;

        }

};

int main()
{
    std::string accountNo;
    std::cout << "Enter you account number: ";
    std::cin >> accountNo;

    int accountPin;
    std::cout << "\nEnter your pin: ";
    std::cin >> accountPin;

    char accountType;
    std::cout << "\nEnter your account type: ";
    std::cin >> accountType;

    bankAccount user1(accountNo, accountPin, accountType);

    float userAmount;
    std::cout << "\nEnter the amount you wish to withdraw: ";
    std::cin >> userAmount;

    user1.withdraw(userAmount);
}
```

Example Code Outputs

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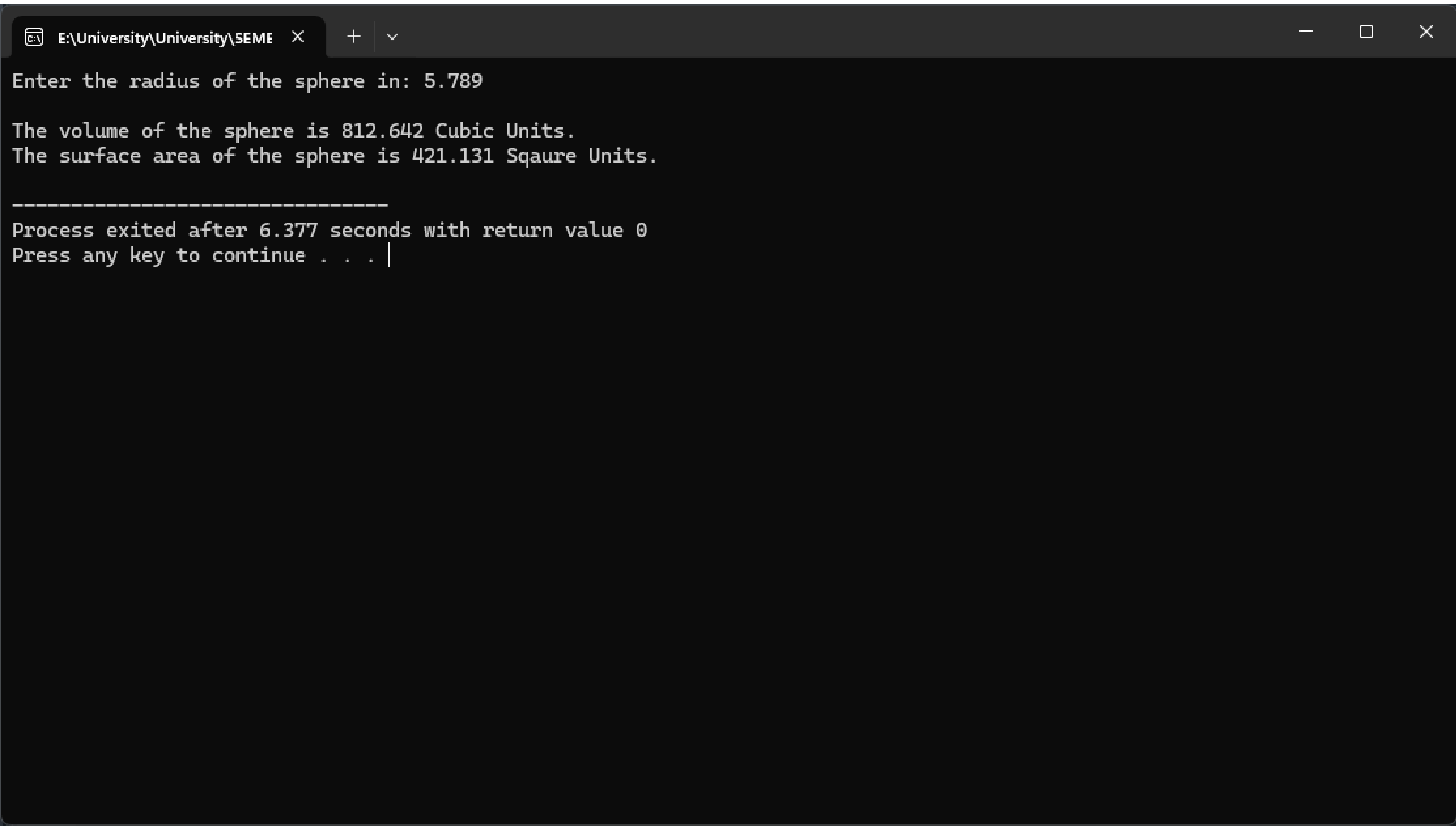
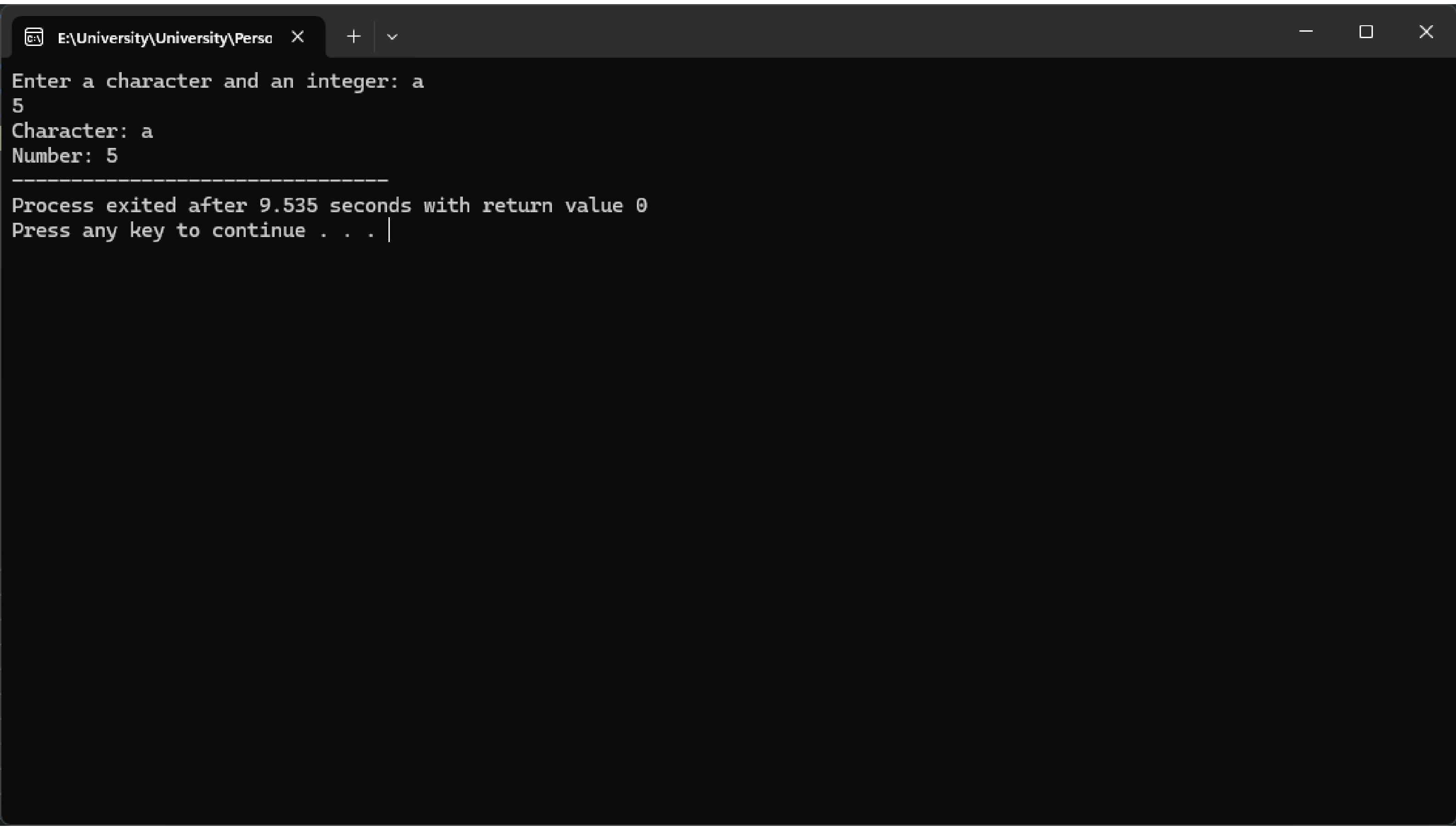
```
Enter you account number: IBN-67341

Enter your pin: 4538

Enter your account type: s

Enter the amount you wish to withdraw: 45900

Transaction successful!
Please, take your card and cash.
Your new balance is 155018
--------------------------------
Process exited after 27.43 seconds with return value 0
Press any key to continue . . .
```


E:\University\University\SEME
Enter the radius of the sphere in: 5.789
The volume of the sphere is 812.642 Cubic Units.
The surface area of the sphere is 421.131 Sqaure Units.

Process exited after 6.377 seconds with return value 0
Press any key to continue . . .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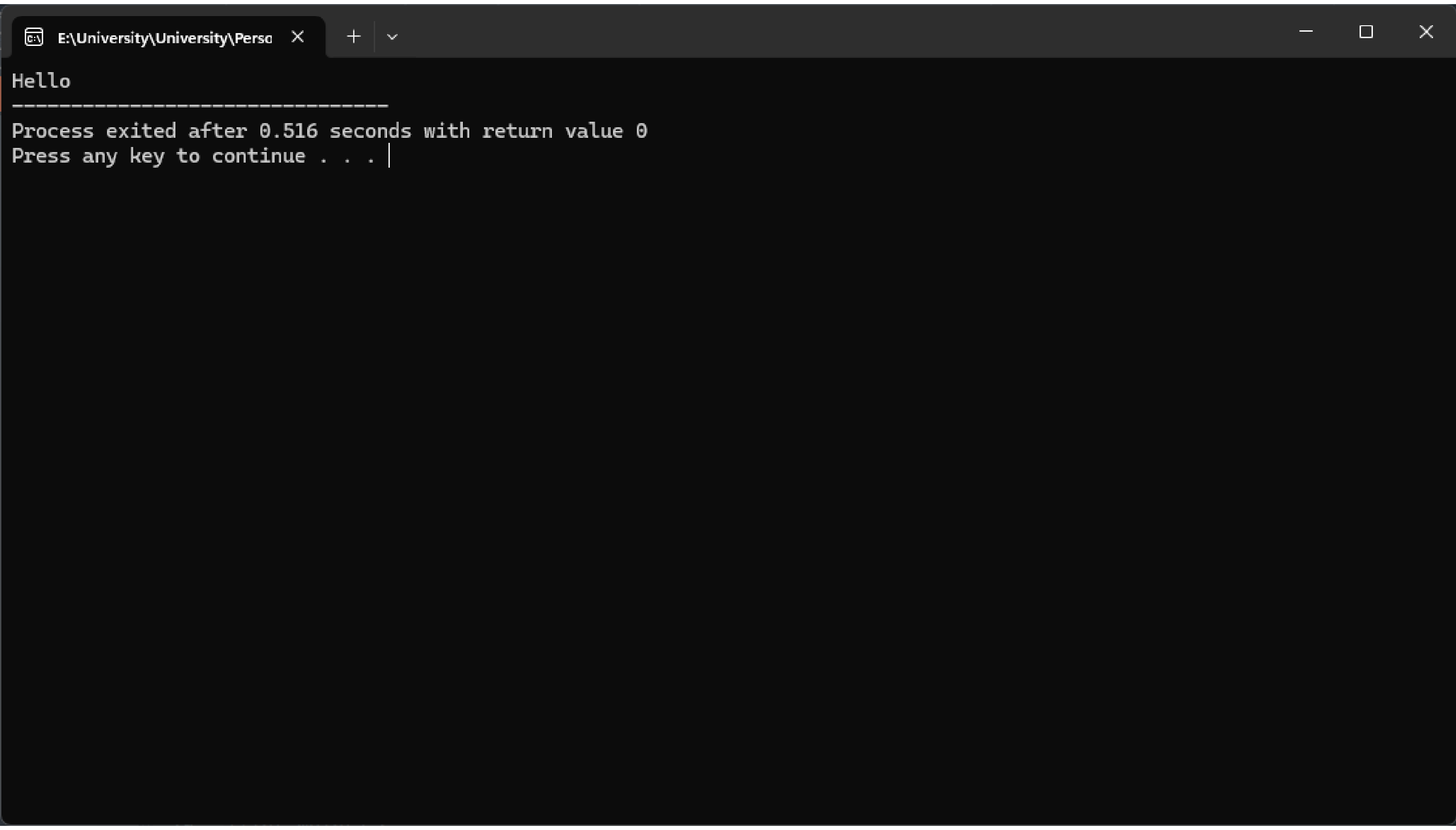
Enter a character and an integer: a
5
Character: a
Number: 5

Process exited after 9.535 seconds with return value 0
Press any key to continue . . .


E:\University\University\Perso
Hello

Process exited after 0.516 seconds with return value 0
Press any key to continue . . .

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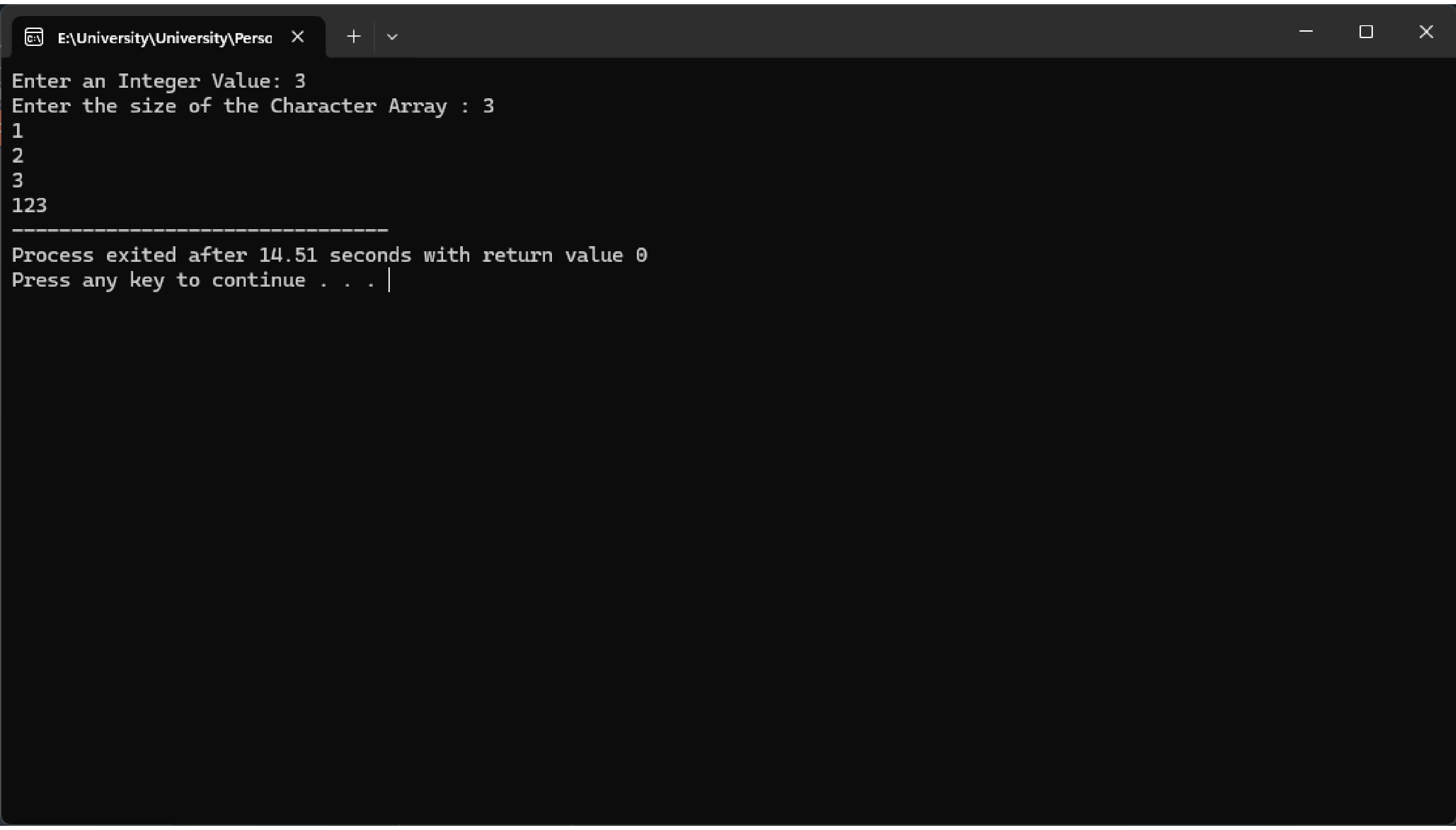
```
Line 1: x = 37
Line 3: *p = 37, x = 37
Line 5: *p = 58, x = 58
Line 6: Address of p = 0x7afe18
Line 7: Value of p = 0x7afe14
Line 8: Value of the memory location pointed to by *p = 58
Line 9: Address of x = 0x7afe14
Line 10: Value of x = 58

--------------------------------
Process exited after 0.4974 seconds with return value 0
Press any key to continue . . .
```


E:\University\University\Perso
Enter an Integer Value: 3
Enter the size of the Character Array : 3
1
2
3
123

Process exited after 14.51 seconds with return value 0
Press any key to continue . . .

```
Line 2: Before calling the function funcValueParam, number = 6
Line 5: In the function funcValueParam, before changing, num = 6
Line 7: In the function funcValueParam, after changing, num = 15
Line 4: After calling the function funcValueParam, number = 6

--------------------------------
Process exited after 0.1947 seconds with return value 0
Press any key to continue . . .
```

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```
Line 1: Based on the course score,
 this program computes the course grade.
Line 4: Enter course score: 79

Line 6: Course score is 79
Line 7: Your grade for the course is C.

--------------------------------
Process exited after 10.5 seconds with return value 0
Press any key to continue . . .
```

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```
Inside test x = 2 and y = 11
Inside test x = 4 and y = 11
Inside test x = 6 and y = 11
Inside test x = 8 and y = 11
Inside test x = 10 and y = 11

--------------------------------
Process exited after 0.505 seconds with return value 0
Press any key to continue . . .
```